PROBLEM 06

Frenet-Serret Frame

Particle on Sinusoidal Curve

Original statement · Problems.pdf, p. 7

A point particle P travels along a curve given by:

$$y = \sin x$$

with a velocity along x given by $\dot{x} = 1$.

(a) Find the intrinsic, unitary vector basis associated to the curve.

(b) Express x as a function of an arc length parameter, s. You can leave your solution as a function of an integral.

(c) Find the tangent and normal acceleration components of P as a function of x.

(d) Write the radius of curvature of the trajectory as a function of x.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle travels along a sinusoidal path $y(x) = \sin x$ with constant horizontal speed $\dot{x} = 1$.
- **Hypotheses:** $\dot{x}(t) \equiv 1\ \text{m/s} \implies x(t) = t + x_0$ (taking $x_0 = 0 \implies x(t) = t$). $\ddot{x} = 0$.
- **Degrees of Freedom:** $\text{CDOF} = 1$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Parametric Curve:** $\mathbf{r}(x) = x\mathbf{i}_0 + \sin x\,\mathbf{j}_0$.
- **Derivative Vector:** $\frac{d\mathbf{r}}{dx} = \mathbf{i}_0 + \cos x\,\mathbf{j}_0$.
- **Differential Arc Length:**

$$ds = \left\| \frac{d\mathbf{r}}{dx} \right\| dx = \sqrt{1 + \cos^2 x}\, dx$$

- **Change of basis (intrinsic basis).** Let $\alpha(x) = \arctan(\cos x)$ be the slope angle of the tangent, so that $\mathbf{e}_t = (\cos\alpha, \sin\alpha) = (1, \cos x)/\sqrt{1+\cos^2 x}$. The in-plane normal used in this solution is $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t = (-\sin\alpha, \cos\alpha) = (-\cos x, 1)/\sqrt{1+\cos^2 x}$, the normal obtained by turning $\mathbf{e}_t$ by $+90^\circ$ (to its left). The columns are the components of $\mathbf{e}_t$, $\mathbf{e}_n$ in $\{\mathbf{i}_0, \mathbf{j}_0\}$:

$$[{}_0R_1] = \frac{1}{\sqrt{1+\cos^2 x}} \begin{pmatrix} 1 & -\cos x \\ \cos x & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \frac{1+\cos^2 x}{1+\cos^2 x} = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- Because $\mathbf{e}_n$ is always the left normal, the sign of a_n tells on which side the centre of curvature lies: $a_n < 0$ (as for $0 < x < \pi$, where $\sin x > 0$) places it to the right of the direction of motion, below the arch; $a_n > 0$ for $\pi < x < 2\pi$ places it above the curve. The matrix properties were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Intrinsic Unitary Vector Basis $\{\mathbf{e}_t, \mathbf{e}_n\}$

- **Unit Tangent Vector $\mathbf{e}_t$:**

$$\mathbf{e}_t = \frac{d\mathbf{r}/dx}{\|d\mathbf{r}/dx\|} = \frac{\mathbf{i}_0 + \cos x\,\mathbf{j}_0}{\sqrt{1+\cos^2 x}}$$

- **Unit Principal Normal Vector $\mathbf{e}_n$:**

In a 2D plane, the normal vector $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t$ is perpendicular to $\mathbf{e}_t$ and oriented to its left (a $+90^\circ$ turn). It points toward the centre of curvature only where the signed curvature is positive.

Differentiating $\mathbf{e}_t$ with respect to arc length s:

$$\frac{d\mathbf{e}_t}{ds} = \frac{d\mathbf{e}_t}{dx}\frac{dx}{ds} = \frac{1}{\sqrt{1+\cos^2 x}}\frac{d}{dx}\left[(1+\cos^2 x)^{-1/2}(\mathbf{i}_0 + \cos x\,\mathbf{j}_0)\right]$$

Applying the quotient/product rule:

$$\frac{d}{dx}\left[\frac{1}{\sqrt{1+\cos^2 x}}\right] = -\frac{1}{2}(1+\cos^2 x)^{-3/2}(2\cos x(-\sin x)) = \frac{\sin x \cos x}{(1+\cos^2 x)^{3/2}}$$

$$\frac{d\mathbf{e}_t}{dx} = \frac{\sin x\cos x}{(1+\cos^2 x)^{3/2}}(\mathbf{i}_0 + \cos x\,\mathbf{j}_0) + \frac{1}{\sqrt{1+\cos^2 x}}(-\sin x\,\mathbf{j}_0)$$

Putting over the common denominator $(1+\cos^2 x)^{3/2}$:

$$= \frac{\sin x\cos x\,\mathbf{i}_0 + \sin x\cos^2 x\,\mathbf{j}_0 - \sin x(1+\cos^2 x)\,\mathbf{j}_0}{(1+\cos^2 x)^{3/2}} = \frac{\sin x\cos x\,\mathbf{i}_0 - \sin x\,\mathbf{j}_0}{(1+\cos^2 x)^{3/2}} = -\frac{\sin x}{(1+\cos^2 x)^{3/2}}(-\cos x\,\mathbf{i}_0 + \mathbf{j}_0)$$

Normalizing to unit length:

$$\mathbf{e}_n = \frac{-\cos x\,\mathbf{i}_0 + \mathbf{j}_0}{\sqrt{1+\cos^2 x}} \quad (\text{left normal } \mathbf{k}_0 \times \mathbf{e}_t)$$

Erratum: The first version called $\mathbf{e}_n$ the normal pointing to the concave side. It is the left normal $\mathbf{k}_0 \times \mathbf{e}_t$ and, since $d\mathbf{e}_t/ds = -\dfrac{\sin x}{(1+\cos^2 x)^{3/2}}\,\mathbf{e}_n$, it points away from the centre of curvature for $0 < x < \pi$ (where $a_n < 0$) and toward it for $\pi < x < 2\pi$ (where $a_n > 0$). Numeric check: sympy gives $a_n(\pi/2) = -1\ \mathrm{m/s^2}$, with the centre of curvature below the crest.

(b) Arc Length Parameter $s(x)$

Pedagogical Justification: The relation between x and arc length s is given by integrating the metric differential:

$$s(x) = \int_0^x \sqrt{1+\cos^2 u}\,du$$

This integral is an incomplete elliptic integral of the second kind, which cannot be expressed in elementary functions and is properly expressed as a quadrature.

(c) Tangent and Normal Acceleration Components

- **Velocity Vector:** Since $\dot{x} = 1$:

$$\mathbf{v} = \dot{x}\mathbf{i}_0 + \dot{y}\mathbf{j}_0 = 1\mathbf{i}_0 + \cos x\,\mathbf{j}_0$$

The speed is:

$$v(x) = \|\mathbf{v}\| = \sqrt{1 + \cos^2 x}$$

- **Acceleration Vector:** Differentiating velocity with $\dot{x} = 1$:

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(1\mathbf{i}_0 + \cos x\, \mathbf{j}_0) = -\sin x\, \dot{x}\, \mathbf{j}_0 = -\sin x\, \mathbf{j}_0$$

- **Tangential Acceleration Component a_t:**

$$a_t = \mathbf{a} \cdot \mathbf{e}_t = (-\sin x\, \mathbf{j}_0) \cdot \left(\frac{\mathbf{i}_0 + \cos x\, \mathbf{j}_0}{\sqrt{1 + \cos^2 x}} \right) = -\frac{\sin x \cos x}{\sqrt{1 + \cos^2 x}}$$

- **Normal Acceleration Component a_n:**

$$a_n = \mathbf{a} \cdot \mathbf{e}_n = (-\sin x\, \mathbf{j}_0) \cdot \left(\frac{-\cos x\, \mathbf{i}_0 + \mathbf{j}_0}{\sqrt{1 + \cos^2 x}} \right) = \frac{-\sin x}{\sqrt{1 + \cos^2 x}}$$

Its magnitude is:

$$|a_n| = \frac{|\sin x|}{\sqrt{1 + \cos^2 x}}$$

(d) Radius of Curvature $\rho(x)$

> **Pedagogical Justification:** Using $a_n = \frac{v^2}{\rho}$:

$$\rho(x) = \frac{v^2}{|a_n|} = \frac{1 + \cos^2 x}{\frac{|\sin x|}{\sqrt{1+\cos^2 x}}} = \frac{(1 + \cos^2 x)^{3/2}}{|\sin x|}$$

(Notice this matches the standard Cartesian curvature formula $\rho = \frac{(1+y'^2)^{3/2}}{|y''|}$ with $y' = \cos x, y'' = -\sin x$!)

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:** the data are in SI units with the length scale 1 m inside the sine ($y = \sin x$ means $y = (1\ \text{m})\sin(x/1\ \text{m})$) and $\dot{x} = 1\ \text{m/s}$. Then $[v] = \text{m/s}$, $[a_t] = [a_n] = \text{m/s}^2$ and $[\rho] = [v^2/|a_n|] = \text{m}$, and $[s] = [\int \sqrt{1+\cos^2 u}\, du] = \text{m}$.
- **Special points:** at the inflection points $x = k\pi$, $\sin x = 0$, so $a_n = 0$ and $\rho \to \infty$ (locally straight). At the crests $x = \pi/2$: $v = 1\ \text{m/s}$, $a_t = 0$, $a_n = -1\ \text{m/s}^2$ and $\rho = v^2/|a_n| = 1$ m, equal to $1/|y''| = 1$ m. At $x = \pi/4$: $\rho = (1+\cos^2 x)^{3/2}/|\sin x| = 3\sqrt{3}/2 = 2.598$ m.
- **Arc length:** $s(\pi) = \int_0^\pi \sqrt{1+\cos^2 u}\, du = 3.8202$ m, greater than the horizontal distance π m, as it must be.
- **Symbolic check (sympy):** $a_t^2 + a_n^2 = \sin^2 x = \|\mathbf{a}\|^2$ and $v^2/|a_n| = (1+\cos^2 x)^{3/2}/|\sin x|$ hold with zero residual.

Source: Problems.pdf (Page 7)